PPG Peak Detectors Evaluation
Report 2

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1 Introduction

A photoplethysmogram (PPG) is a small device used to measure changes in the volume of veins in the human body. This study presents an evaluation of three peak detectors applied to photoplethysmographic (PPG) signals taken from the TROIKA dataset. In this work, some PPG peaks detectors are evaluated assessing the performance of them by comparing the number of detected peaks in different PPG signals. Another comparison is done using the ECG registration and the accelerometer, to check if the human is walking or running.

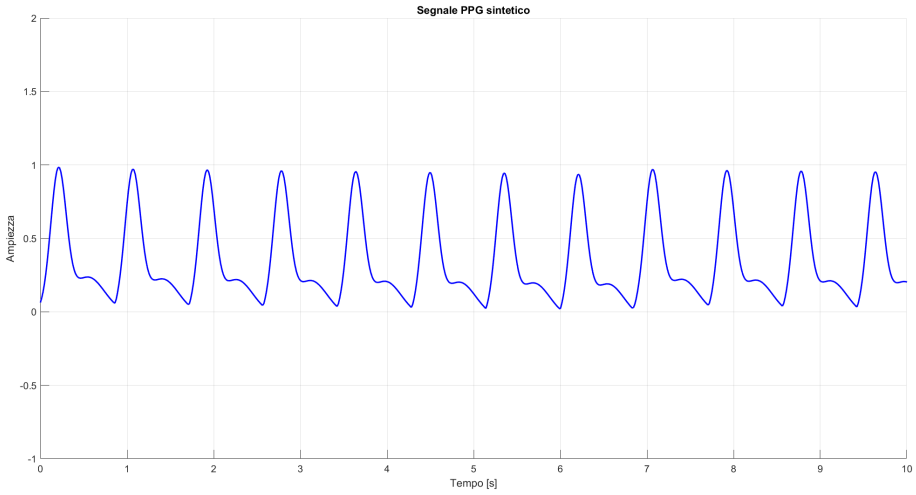


Figure 1.1: Example of PPG signal

2 Dataset - TROIKA

The dataset used in this study comes from the TROIKA framework, which provides PPG signals, ECG reference signals, and accelerometer data. The Troika dataset is used to study heart rate measurement from wearable devices. It has data from 12 people who ran at different speeds. It helps to test how well heart rate can be measured during movement. All signals were sampled at 125 Hz and the analysis was performed on the first 30 seconds and the full 30-minute traces from two recordings found into "DATA 11 TYPE02.mat".

3 Peak detectors

In this study, three different PPG peak detectors were implemented and evaluated:

- **AMPD**, designed to find peaks in periodic or almost-periodic signals such as PPG. It uses a multiscale analysis of local maxima;
- **MSPTD**, which detects both peaks and troughs using a Local Maxima Scalogram (LMS) at different scales;
- **qppgfast**, which uses the slopes of the PPG signal to detect systolic upslopes and applies an adaptive threshold to find peaks quickly.

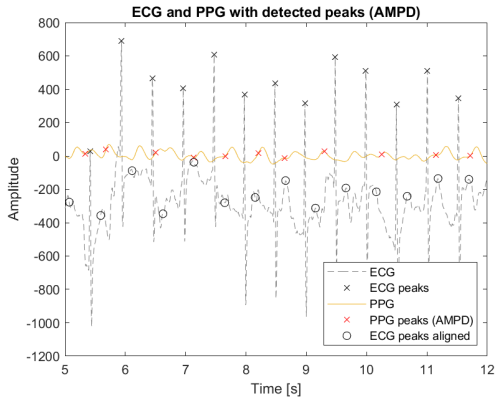
4 Results

DATA_11_TYPE02.mat				
Peak Detector	**First 30s**		**Full Track**	
	First Trace	**Second Trace**	**First Trace**	**Second Trace**
ECG	56	56	613	634
qppgfast	55	55	674	674
MSPTD	50	50	678	678
AMPD	49	49	673	673

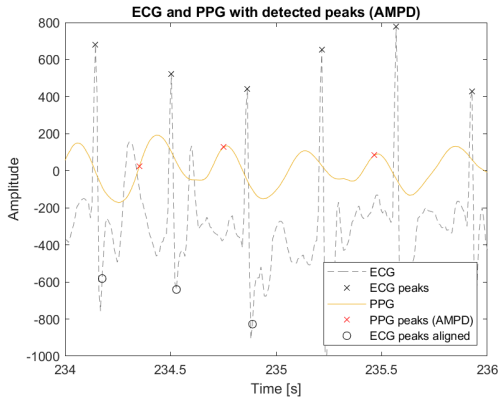
Table 1: Number of peaks detected by each algorithm in the two traces of DATA_11_TYPE02.mat

The peak counts summarized in Table 1 show how each detector performed on the PPG signals compared to the reference ECG. In the first 30 seconds, before the subject starts running, the number of PPG peaks detected is very close to the number of heartbeats in the ECG (56 beats for both PPG traces). For example, qppgfast found 55 peaks in this interval, missing only one beat, while MSPTD and AMPD found 50 and 49 peaks, respectively. These small differences indicate that a few PPG pulses were missed by MSPTD and AMPD even in a almost static segment. The full recording (during which the subject is running and moving a lot), all three algorithms report more peaks than the ECG's true count. For example qppgfast counted 674 vs 613 ECG beats on Trace 1. This means the detectors registered many false peaks due to motion artifacts. The accelerometer's variations in the full track are much higher, indicating strong movement. This motion caused noisy fluctuations in the PPG waveform that the algorithms identified as heartbeats.

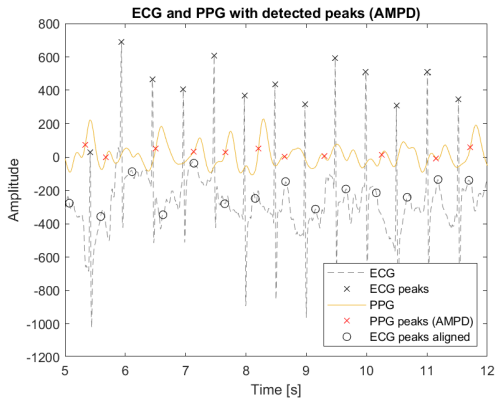
4.1 AMPD



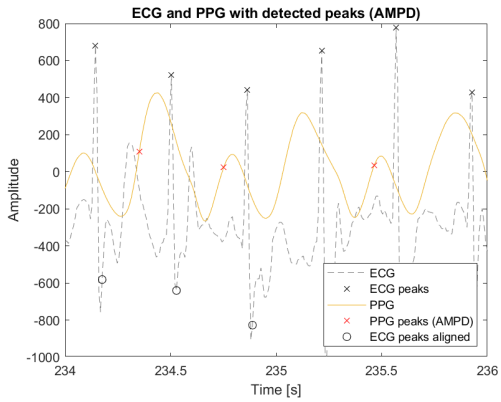
4.1 First 30 seconds of Trace 1



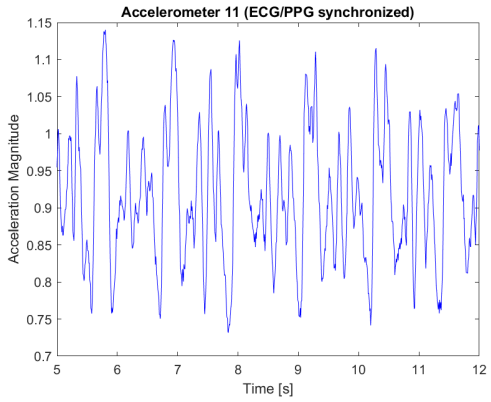
4.2 Full track of Trace 1



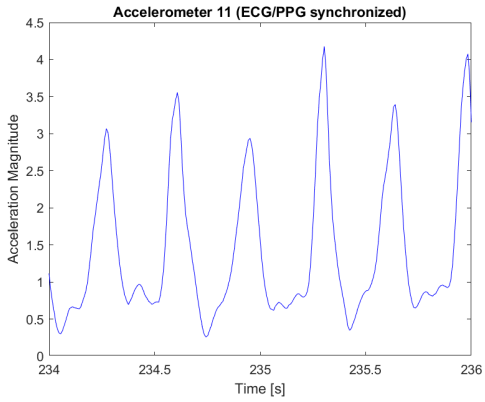
4.3 First 30 seconds of Trace 2



4.4 Full track of Trace 2



4.5 Accelerometer 5-12 sec



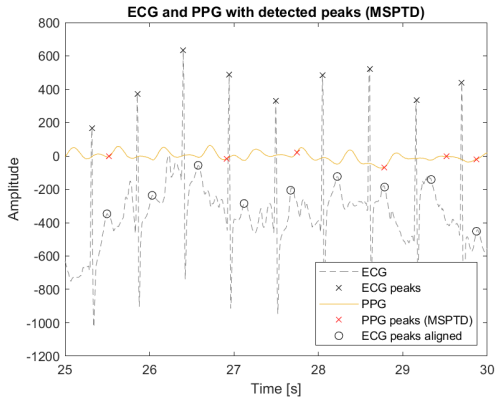
4.6 Accelerometer 234-236 sec

Figure 4.1: AMPD results

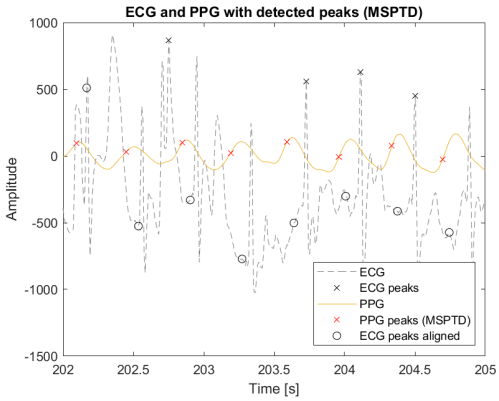
In the first 30 seconds of Trace 1 (Figure 4.1), the AMPD algorithm detects most of the PPG peaks, but it misses a few beats. The reference ECG for this period shows about 56 true beats in the first 30 seconds, while AMPD finds roughly 49 peaks in the same interval, indicating that some heartbeats were not detected. These missed peaks might be caused by the algorithm's sensitivity to small or flat PPG pulses rather than motion artifacts, as the accelerometer data (Figure 4.5) shows dynamic movement with a lot of noise. A similar result is seen in the first 30 seconds of Trace 2, where AMPD also misses some peaks, again probably due to low-amplitude pulses.

During the motion period (Figures 4.2 and 4.4) after 30 seconds of recording, AMPD shows a large increase in the number of detected peaks compared to the true ECG beats (673 vs 613 for Trace 1 and 673 vs 634 for Trace 2). This means that AMPD produced many false positives, probably because motion artifacts in the PPG signal caused the algorithm to detect noise instead of real peaks. The accelerometer signals in Figure 4.6 confirm some movement, explaining the false detections. In particular in the Figure 4.4, the missed peak between 235 and 235.5 is syncronized with the biggest acceleration magnitude of the window. AMPD worked well but sometimes made mistakes. It often found too many peaks instead of missing one. This caused many false positives.

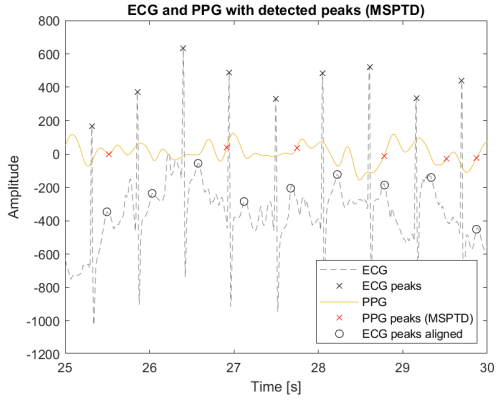
4.2 MSPTD



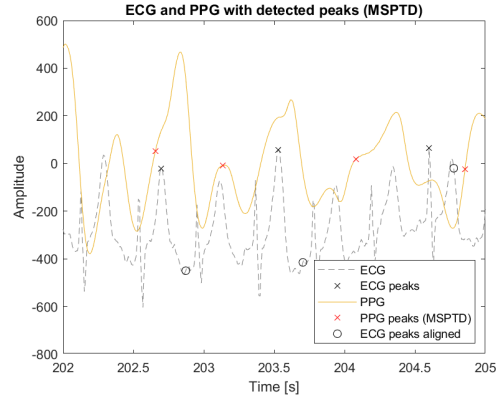
4.7 First 30 seconds of Trace 1



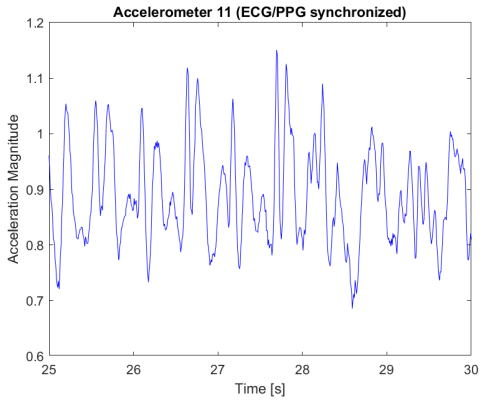
4.8 Full track of Trace 1



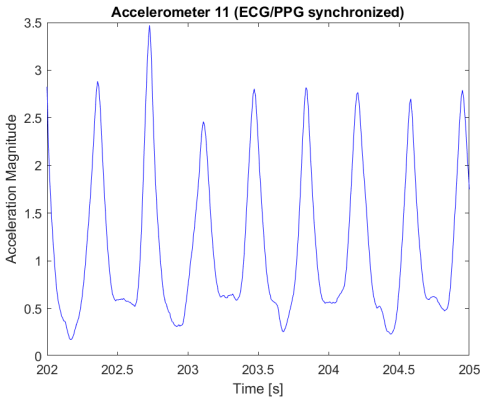
4.9 First 30 seconds of Trace 2



4.10 Full track of Trace 2



4.11 Accelerometer 25-30 sec



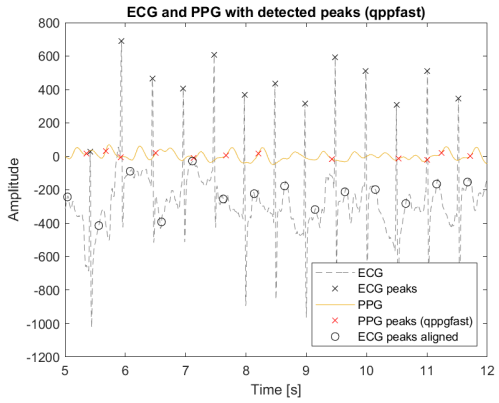
4.12 Accelerometer 202-205 sec

Figure 4.2: MSPTD results

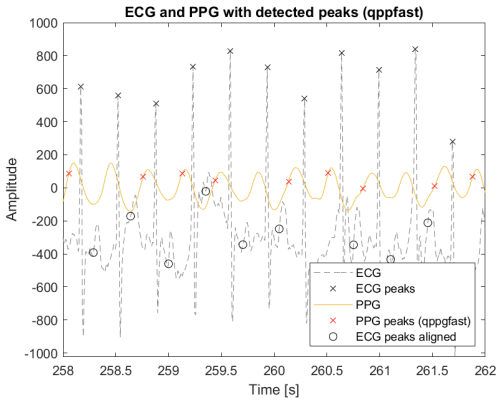
At the start of Trace 1 (Figure 4.7), MSPTD detects most peaks but still misses some heartbeats. The missed peaks are probably due to low-amplitude pulses that the algorithm could not detect. The accelerometer signal in Figure 4.11 shows motion, suggesting that motion artifacts could also be contributing to these detection errors. Similar performance is seen in Trace 2 during the first 30 seconds, with MSPTD missing a few small peaks at the same time.

In the motion period (Figures 4.8 and 4.10), MSPTD shows a big increase in the number of detected peaks (Table 1). This means MSPTD was heavily affected by motion artifacts, which caused false detections. Actually the accelerometer signals in Figures 4.12 shows a periodic and regular pattern with clear peaks, indicating a rhythmic movement, The acceleration values are higher which means the movements are stronger. This interval of the accelerometer is cleaner with less noise.

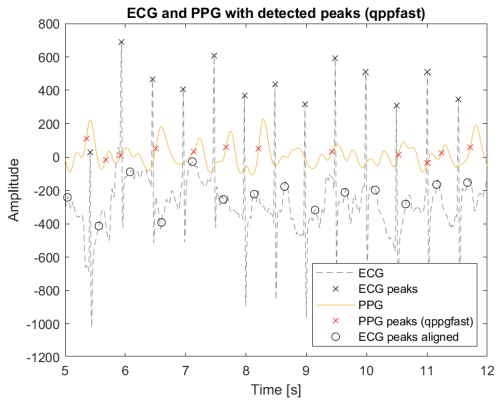
4.3 qppgfast



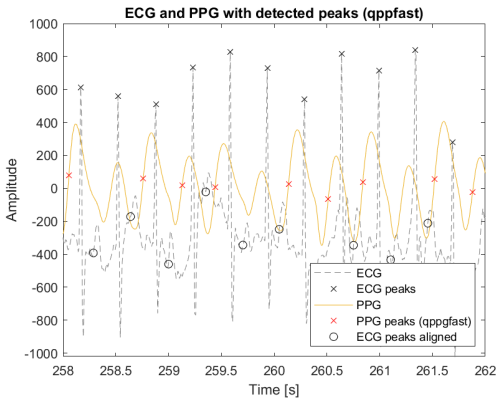
4.13 First 30 seconds of Trace 1



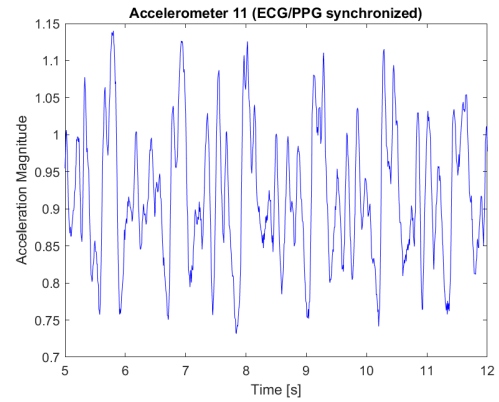
4.14 Full track of Trace 1



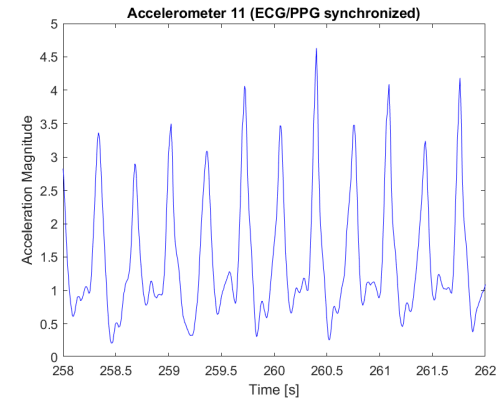
4.15 First 30 seconds of Trace 2



4.16 Full track of Trace 2



4.17 Accelerometer 5-12 sec



4.18 Accelerometer 258-262 sec

Figure 4.3: qppgfast results

qppgfast performs well, detecting almost every peak. Even this detector found more peaks than the ECG trace. The Table 1 shows only one missed peak in the first 30 seconds while the performance in the rest of the traces is almost the same. The ECG trace is more or less clear, so the cause of the mistakes could be the accelerometer in Figure 4.18. The magnitude has high peaks causing miss peaks by the detector in the interval shown, confirming that these extra peaks come from noise rather than real heartbeats. Hence, qppgfast performs best when the signal is clean, but it has some troubles during movement, producing many false peaks.

5 Conclusion

All three PPG peak detectors (AMPD, MSPTD, and qppgfast) perform well in clean and static conditions, despite the accelerometer's trace. During the first 30 seconds, each detector finds almost the same number of peaks as the ECG, with qppgfast missing the fewest beats. AMPD and MSPTD miss more than 10% of the peaks. This shows that qppgfast is the most sensitive detector when there is no motion. However, all three detectors are quite accurate under these conditions.

When the subject starts moving, all detectors produce many false positives. MSPTD shows the largest number of false detections during motion with 678 peaks detected, causing an error of about 11% for the first trace and 7% for the second. The accelerometer confirms that motion artifacts cause these errors, as the detectors cannot always separate real heartbeats from noise caused by movement. This study shows that even with many false positives, real beats can still be missed if the PPG signal is too noisy or too flat.

There are only small differences between the two traces, and they show a similar shape. The PPG peak detectors found the same number of peaks, probably because the PPG sensors were placed very close to each other while recording the ECG trace.